

DS	Session	Searching	Question Information	Level 1	Challenge 6
Problem Description: Kanna is upset to learn that no one at his school recognises his first name. Even his friends refer to him by his surname. Frustrated, he decides to make his fellow college students know his first name by forcing them to solve this question. The task is determining the third greatest number in the supplied array. Constraints: $0 < n < 100$ $0 < arr < 1000$ Input Format: first line represents the number of elements N to be get second line indicates input elements according to N Output Format: Single line represents the out put that is third largest number.					

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

```
6
1 14 2 16 10 20
```

EXPECTED OUTPUT

The third Largest element is 14

Test Case 2

INPUT (STDIN)

```
7
19 -10 20 14 2 16 10
```

EXPECTED OUTPUT

The third Largest element is 16

▼ Mandatory Test Cases

Test Case 1

KEYWORD

```
void thirdLargest(int
arr[],int arr_size)
```

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

7

Test Case 2

TOKEN COUNT

220

Test Case 3

NLOC

44

Searching6.c +

```
1 #include <stdio.h>
2 #include <limits.h>
3
4 void thirdLargest(int arr[], int arr_size) {
5     if (arr_size < 3) {
6         return;
7     }
8     int first = INT_MIN, second = INT_MIN, third = INT_MIN;
9     for (int i = 0; i < arr_size; i++) {
10         if (arr[i] > first) {
11             third = second;
12             second = first;
13             first = arr[i];
14         } else if (arr[i] > second && arr[i] != first) {
15             third = second;
16             second = arr[i];
17         } else if (arr[i] > third && arr[i] != second && arr[i] != first) {
18
19         }
20     }
21     printf("The third Largest element is %d\n", third);
22 }
23
24 int main() {
25     int n;
```

main | CH.SC.U4CSE25225 | Searching6.c

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```
26     if (scanf("%d", &n) != 1) return 0;
27     int arr[1000];
28     for (int i = 0; i < n; i++) {
29         scanf("%d", &arr[i]);
30     }
```

```
31  thirdLargest(arr, n);  
32  return 0;  
33}  
34
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
6
```

```
1 14 2 16 10 20
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching6.exe
```

```
The third Largest element is 14
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
7
```

```
19 -10 20 14 2 16 10
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching6.exe
```

```
The third Largest element is 16
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```